So, network address for 63rd subnet 2 is 192.168.1.~~64~~

SUBNETTING - PART (II)

Day-14

SUBNETTING TRICK.

192.168.1.0/26

Network | Host

11000000 . 10101000 . 00000001 . 00 | 000000

128	64	32	16	8	4	2	1
0	0	0	0	0	0	0	0

Last bit of network portion is 64. This means to find next subnet we just have to add 64..

192.168.1.64/26 → network address of subnet 2.

192.168.1.~~64~~128/26 → " " of subnet 3

192.168.1.192/26 → " " of subnet 4.

Scanned with CamScanner

2^x = no. of subnet
x = no. of 'borrowed bits'

Divide the 192.168.255.0/24 network into five subnets of equal size. Identify the five subnets.

We just need to find how many bits we have to borrow from host portion

Note:
2^x = no. of subnet, where x = no. of 'borrowed bits'
$2^n - 2$ = no. of usable hosts, n = no. of host bits

We need 5 subnets, $2^x = 5$, its not always possible to match the no. of subnets or no. of hosts we need.
$2^3 = 8$, we need 5. $2^2 = 4$. So, if we use borrow 2 bits it won't be enough. So, we need to borrow 3 bits. i.e /27 netmask.

So, our subnets will be as follow:
1st Subnet = 192.168.1.0/27
2nd Subnet = 192.168.1.32/27
3rd subnet = 192.168.1.64/27
4th subnet = 192.168.1.96/27
5th subnet = 192.168.1.128/27

Though our /27 prefix length allows upto 8, maile 5 ota no matta lekhyo kos we need 5.

Identify the subnet.

What subnet does host 192.168.5.57/27 belong to?
192.168.5.57 → 11000000.10101000.00000101.00111001
To find network address change host bit to 0, (if broadcast change to
Nov. 4th

Now last octet is
001 00000.
So, 192.168.5.57 belongs to 192.168.5.32/27 subnet

What subnet does host 192.168.29.219/29 belong to?
Subnet ID : 192.168.29.216/29

Subnet / Hosts (Class C)

Prefix Length	No. of Subnets	No. of hosts
/25	2	126
/26	4	62
/27	8	30
/28	16	14
/29	32	6
/30	64	2
/31	128	0(2)
/32	256	0(1)

NOTE:
The process of subnetting Class A, class B & class C network is exactly same.

Q) You have been given the 172.16.0.0/16 network. You are asked to create 80 subnets for your company's various LANs. What prefix length should you use?
64 < 80 < 128. ie $2^6 < 80 < 2^7$.
So, we'll need 128 subnets, from which we'll use 80 subnets

So, the prefix length is /23.
172.16.0.0/23.

Some subnets are; 172.16.0.0/23, 172.16.2.0/23, 172.16.4.0/23 - - - .

Q.2) "... 172.22.0.0/16. You are req. to divide network into 500 separate subnets. What prefix length should you use.

to $2^9 = 512$.

We can borrow bits from last octet too, so we can use /25, /26 also.

So prefix length is /25.

You've been given 172.18.0.0/16 network. Your company requires 250 subnets with same no. of hosts per subnet. What prefix length should you use?

Borrowing 8 bits = 256 subnets

8 host bits = 254 hosts per subnet

What subnet does host 172.25.217.192/21 belong to?

b) 172.25.216.0/21

QUIZ

a) You've been given the 172.30.0.0/16 network. Your company requires 100 subnets with at least 500 hosts per subnet. What prefix length you should use?

$2^6 < 100 < 2^7$. So, we'll burrow 7 bits

After taking 7 bits, no. of remaining host bits = 9.

$2^9 - 2 = 510$ usable address.

So, prefix length is 172.30.0.0/23

Q-2) What subnet does host 172.21.111.201/20 belong to?

172.21.2111.201/20

10101100.00010101.01100000.00000000

172.21.96.0 /20

3) What is the broadcast address of network 192.168.91.78/26 belongs to?

11000000.10101000.01011011.01001110

192 . 168 . 91 . 78.

11000000.10101000.01011011.01111111

Broadcast address: 192.168.91.127/26

4). You divide the 172.16.0.0/16 network in 4 no subnets of equal size. Identify network & broadcast of second subnet.

Prefix length = /18 as $2^2 = 4$ subnet

172. 16.0. 0.

10101100.00010000.00000000.00000000

172. 16 . 64 . 0 → Network addr of second subnet

10101100.00010000.01000000.00000000

172.16.127.255 → broadcast of subnet 2

6) You divide the 172.30.0.0/16 into subnets of 1000 hosts each. How many subnets are you able to make?

For 10,000 hosts we need 10 host bits.
So, we get 6 borrowed bits. which means we can make 64 subnets.

Subnetting class A Networks (Day-15)

7 You have been given the 10.0.0.0/8 network. You must create 2000 subnets which will distributed to varias enterprises. What prefix length must you use?